ONE REAL PLANE, MANY COMPLEX STRUCTURES
EXPLICIT COMPARISONS, LATTICES, AND COMPLEX TORI

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Abstract. A complex structure on a real plane is an operator whose square is minus the identity. We classify the structures compatible with a fixed orientation by the upper half-plane, compute their comparison maps, and determine when these maps descend to quotients by lattices. We prove the affine description of all holomorphic maps between one-dimensional complex tori and derive the modular equivalence criterion. The tori with parameters i and $2i$ give an explicit example of a common smooth underlying space, distinct complex structures, and degree-two isogenies. A final discussion identifies the limited role of these classical constructions as an entry point to inter-universal Teichmüller theory. All exercises include solutions.

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1. Scope and the objects being compared

The basic question is precise: if two objects have the same underlying real space, what further conditions make a map between them preserve their complex structures? Passing to a lattice quotient introduces an additional condition that is independent of complex linearity. Both distinctions can be studied by explicit matrices.

We use column vectors and the standard oriented basis e_1, e_2 of $V = \mathbb{R}^2$. Throughout, a complex structure is *constant* on this vector space. On its quotients we use the translation-invariant complex structures induced by these constant operators. We do not classify arbitrary almost complex structures on higher-dimensional manifolds.

The prerequisites are elementary real and complex linear algebra, the definition of a holomorphic function, Liouville's theorem, and basic covering space theory. The two analytic ingredients in the classification of maps of tori will be identified explicitly. No IUT construction or assertion about Diophantine inequalities is used in the proofs.

2. A complex structure is an additional operation

Definition 2.1. A *complex structure* on a real vector space V is a real linear map $J : V \to V$ satisfying $J^2 = -\operatorname{id}_V$. It defines complex scalar multiplication by

$$(a + ib) \cdot_J v = av + bJv, \qquad a, b \in \mathbb{R}. \tag{2.1}$$

This really is a complex vector space structure. For example,

$$(a \operatorname{id} + bJ)(c \operatorname{id} + dJ) = (ac - bd) \operatorname{id} + (ad + bc)J,$$

which is exactly the rule for multiplying $a + ib$ and $c + id$; the other scalar axioms follow from real linearity. Conversely, multiplication by i in a complex vector space is such an operator. Thus specifying J and specifying an extension of the given real scalar multiplication to complex scalars are equivalent data.

In real dimension two, a nonzero vector v and Jv are linearly independent: otherwise $Jv = rv$ for some real r, and applying J gives $-v = r^2 v$. Consequently (V, J) has complex dimension one. The ordered basis (v, Jv) determines an orientation independent of v. Indeed, if $w = av + bJv \neq 0$, then

$$[\, w \;\; Jw \,] = [\, v \;\; Jv \,] \begin{pmatrix} a & -b \\ b & a \end{pmatrix},$$

and the latter matrix has determinant $a^2 + b^2 > 0$. We call J *positive* if this orientation is the fixed standard one.

Proposition 2.2. *A real linear map $B : (V, J) \to (W, K)$ is complex linear if and only if*

$$BJ = KB. \tag{2.2}$$

If B is invertible, this is equivalent to $K = BJB^{-1}$.

Proof. Complex linearity implies preservation of multiplication by i, hence $B(Jv) = K(Bv)$. Conversely, this identity and real linearity give

$$B(av + bJv) = aBv + bKBv = (a + ib) \cdot_K Bv.$$

The assertion for invertible maps follows by multiplying on the right by B^{-1}. □

Every nonzero complex linear map between complex lines is invertible. It preserves their complex orientations: in complex bases its real matrix has the form

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}, \qquad a + ib \neq 0.$$

In particular, a complex linear isomorphism between two positive structures on the fixed oriented plane has positive real determinant.

3. An explicit family and two different comparison maps

Set

$$J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad A_\lambda = \begin{pmatrix} \lambda & 0 \\ 0 & 1 \end{pmatrix}, \qquad \lambda > 0.$$

The transported structure is

$$J_\lambda = A_\lambda J_0 A_\lambda^{-1} = \begin{pmatrix} 0 & -\lambda \\ \lambda^{-1} & 0 \end{pmatrix}. \tag{3.1}$$

Matrix multiplication gives $J_\lambda^2 = -I$. It is positive because $\det[e_1, J_\lambda e_1] = \lambda^{-1} > 0$.

There are now *two* maps to distinguish. The identity on the underlying real plane is a real linear isomorphism

$$\mathrm{id} : (V, J_0) \longrightarrow (V, J_\lambda),$$

but Proposition 2.2 says it is complex linear exactly when $J_0 = J_\lambda$, that is, when $\lambda = 1$. On the other hand,

$$A_\lambda : (V, J_0) \longrightarrow (V, J_\lambda)$$

is a complex linear isomorphism for every $\lambda > 0$.

These statements can also be checked without matrix terminology. Put

$$F_0(x, y) = x + iy, \qquad F_\lambda(x, y) = x + i\lambda y.$$

Both maps are real linear isomorphisms to $\mathbb{C}$, and

$$F_0 J_0 = iF_0, \qquad F_\lambda J_\lambda = iF_\lambda.$$

The coordinate expression of A_λ is

$$F_\lambda A_\lambda F_0^{-1}(z) = \lambda z, \tag{3.2}$$

which is complex linear. The coordinate expression of the underlying identity is instead

$$F_\lambda F_0^{-1}(z) = \frac{1+\lambda}{2} z + \frac{1-\lambda}{2} \bar{z}. \tag{3.3}$$

This follows from $x = (z + \bar{z})/2$ and $y = (z - \bar{z})/(2i)$. The antiholomorphic term vanishes exactly when $\lambda = 1$.

Example 3.1 (The first nontrivial case)**.** For $\lambda = 2$,

$$J_2 e_1 = \tfrac{1}{2} e_2, \qquad J_2 e_2 = -2e_1.$$

The identity cannot preserve multiplication by i, since it sends $J_0 e_1 = e_2$ to e_2, whereas $J_2\,\mathrm{id}(e_1) = e_2/2$. The map A_2 does preserve it: $A_2 J_0 e_1 = e_2 = J_2(2e_1) = J_2 A_2 e_1$.

Complex structures do not specify a metric. For example, A_λ does not preserve the fixed Euclidean metric unless $\lambda = 1$. If we wish to use a metric compatible with J_λ, we may choose $g_\lambda = dx^2 + \lambda^2 dy^2$, the pullback of the Euclidean metric by F_λ. Direct substitution shows $g_\lambda(J_\lambda v, J_\lambda w) = g_\lambda(v, w)$. This is an additional choice, not part of the definition of J_λ.

4. All positive structures and the upper half-plane

The family J_λ covers only a one-real-parameter subset. The complete parameter space has two real dimensions.

Theorem 4.1. *Let* $\mathbb{H} = \{\tau = u + iv \in \mathbb{C} : v > 0\}$. *The assignment*

$$\tau \longmapsto J_\tau = \frac{1}{v}\begin{pmatrix} -u & -(u^2+v^2) \\ 1 & u \end{pmatrix} \tag{4.1}$$

is a bijection from $\mathbb{H}$ *to the positive complex structures on the fixed oriented plane with basis* e_1, e_2. *Its complex coordinate is*

$$F_\tau(x,y) = x + \tau y, \qquad F_\tau J_\tau = iF_\tau. \tag{4.2}$$

In particular, $J_{i\lambda} = J_\lambda$.

Proof. The determinant of the real coordinate map $F_\tau : V \to \mathbb{C}$ is $v > 0$, so it is invertible and orientation preserving. Solving $x' + \tau y' = i(x + \tau y)$ gives

$$y' = \frac{x+uy}{v}, \qquad x' = -\frac{ux + (u^2+v^2)y}{v}.$$

This is precisely the matrix in (4.1). It is the pullback of multiplication by i under F_τ, so it squares to $-I$ and is positive.

Conversely, let J be positive. Write its matrix as $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Since (e_1, Je_1) is positively oriented, $c > 0$. From the lower-left entry of $J^2 = -I$ we obtain $c(a+d) = 0$, hence $d = -a$. The upper-left entry then gives $a^2 + bc = -1$. Define

$$v = \frac{1}{c} > 0, \qquad u = -\frac{a}{c}.$$

It follows that $a = -u/v$, $d = u/v$, and $b = -(1+a^2)/c = -(u^2+v^2)/v$. Thus $J = J_{u+iv}$. The same formulas recover u, v uniquely from J. □

There is also a coordinate-free quotient description. Every positive J is conjugate to J_0 by an element of $\mathrm{GL}^+(2,\mathbb{R})$, the group of real matrices with positive determinant: take the matrix with columns w, Jw. It intertwines J_0 and J. The stabilizer of J_0 is computed by solving $BJ_0 = J_0B$:

$$B = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}, \qquad a^2 + b^2 > 0.$$

This is the group $\mathbb{C}^\times$ acting by complex multiplication. Therefore the orbit map gives

$$\{\text{positive complex structures on } V\} \simeq \mathrm{GL}^+(2,\mathbb{R})/\mathbb{C}^\times \simeq \mathbb{H}. \tag{4.3}$$

Here the first quotient is a quotient by the stabilizer, not a quotient by all real linear changes of coordinates. In fact, *all* these structures are isomorphic as unmarked complex vector spaces. The parameter τ records their placement on a *fixed* real plane with a fixed basis.

For later calculations, all real linear maps between two such planes have the explicit form

$$B = F_\sigma^{-1}(\alpha z + \beta \bar{z})F_\tau, \qquad \alpha, \beta \in \mathbb{C},$$

where the middle expression denotes a real linear operator on $\mathbb{C}$. The map is complex linear precisely when $\beta = 0$. Its real determinant is

$$\det_{\mathbb{R}} B = \frac{\operatorname{Im}\tau}{\operatorname{Im}\sigma}(|\alpha|^2 - |\beta|^2).$$

To verify the middle determinant formula, write $\alpha = a + ib$, $\beta = c + id$; the real matrix is $\left(\begin{smallmatrix} a+c & -b+d \\ b+d & a-c \end{smallmatrix}\right)$, whose determinant is $a^2 + b^2 - c^2 - d^2$.

5. A lattice introduces a second preservation condition

Definition 5.1. A *lattice* in $\mathbb{R}^2$ is a subgroup $L = \mathbb{Z}\ell_1 + \mathbb{Z}\ell_2$ generated by a real basis. The quotient V/L is a smooth real torus. For a constant complex structure J, translations have derivative I and preserve J. Local complex coordinates on V therefore induce a complex structure on V/L.

More explicitly, a sufficiently small open disc in V has disjoint translates by nonzero lattice elements, and its projection to V/L is a coordinate neighborhood. On overlaps, the lifted coordinates differ by translations, which are holomorphic. This constructs the complex atlas without any integrability theorem. Compactness follows from the compact closed fundamental parallelogram.

For $L = \mathbb{Z}^2$ and $J = J_\tau$, the map F_τ identifies this quotient with

$$E_\tau = \mathbb{C}/\Lambda_\tau, \qquad \Lambda_\tau = \mathbb{Z} + \tau\mathbb{Z}. \tag{5.1}$$

We work with complex tori as complex analytic groups. Their identification with algebraic elliptic curves is a further classical theorem and is not needed for the following arguments.

Proposition 5.2 (Descent of a real linear map)**.** *Let $B : V \to V'$ be real linear and let L, L' be lattices. The formula*

$$\overline{B} : V/L \longrightarrow V'/L', \qquad [v] \longmapsto [Bv]$$

is well-defined if and only if $B(L) \subseteq L'$. When it is well-defined, it is holomorphic for the structures J, K if and only if $BJ = KB$. If B is invertible, the induced map is a covering of degree $[L' : B(L)]$, and is an isomorphism exactly when $B(L) = L'$.

Proof. Two representatives differ by $\ell \in L$. Their images differ by $B\ell$, so the required independence of representatives is equivalent to $B(L) \subseteq L'$. In lifted complex coordinates, the differential is B at every point. Its complex linearity is exactly (2.2). For sufficiency this gives a complex linear local coordinate formula; for necessity use the complex linear differential of a holomorphic map.

When B is invertible, it is a local diffeomorphism. It is surjective on quotients because any $v' \in V'$ has a preimage $B^{-1}v'$. Its kernel is $B^{-1}(L')/L$, identified by B with $L'/B(L)$. This finite group has order $[L' : B(L)]$; all fibers are its translates. To see finiteness directly, translate each point of L' by an element of $B(L)$ into a bounded fundamental parallelogram for $B(L)$. Its closure meets the discrete lattice L' in finitely many points, supplying finitely many coset representatives. The local diffeomorphism and these descriptions give the claimed covering. It is bijective exactly when this index is one, in which case B^{-1} also descends and supplies an inverse. □

For the fixed lattice $\mathbb{Z}^2$, an invertible real matrix descends precisely when all its entries are integers; it gives a real torus automorphism precisely when it lies in $\mathrm{GL}(2,\mathbb{Z})$. If it is also complex linear between positive structures, its determinant is positive, and the automorphism condition becomes $B \in \mathrm{SL}(2,\mathbb{Z})$. Thus complex linearity and preservation of the lattice are distinct tests.

Example 5.3 (Transporting the lattice or fixing it)**.** For any $\lambda > 0$, A_λ gives a holomorphic isomorphism

$$(V/\mathbb{Z}^2, J_0) \longrightarrow (V/A_\lambda\mathbb{Z}^2, J_\lambda).$$

In target complex coordinates the lattice is $F_\lambda(A_\lambda\mathbb{Z}^2) = \lambda(\mathbb{Z} + i\mathbb{Z})$, so this map is just $[z] \mapsto [\lambda z]$ between homothetic lattices.

If the target lattice is instead kept equal to $\mathbb{Z}^2$, A_λ descends only when λ is a positive integer. For $\lambda = n$ the map is

$$E_i \longrightarrow E_{ni}, \qquad [z] \longmapsto [nz],$$

of degree n, because $\det A_n = n$. For $n > 1$, its inverse on the plane does not descend in the reverse direction.

6. All holomorphic maps of one-dimensional complex tori

The descent proposition starts with linear maps. We now prove that there are no additional origin-preserving holomorphic maps to consider.

Theorem 6.1 (Affine lifting)**.** *Let $\Lambda, \Lambda' \subset \mathbb{C}$ be lattices. Every holomorphic map $f : \mathbb{C}/\Lambda \to \mathbb{C}/\Lambda'$ is induced by*

$$F(z) = cz + t, \qquad c, t \in \mathbb{C}, \quad c\Lambda \subseteq \Lambda'. \tag{6.1}$$

Conversely, every such formula defines a holomorphic map. The constant c is uniquely determined by f, and t is determined modulo Λ'. If $f(0) = 0$, one can choose $t = 0$, and f is a group homomorphism.

Proof. Write q, q' for the quotient maps. The map $f \circ q : \mathbb{C} \to \mathbb{C}/\Lambda'$ has a lift $F : \mathbb{C} \to \mathbb{C}$ under q', after choosing $F(0)$. Here we use the lifting theorem for coverings: a map from a simply connected, locally path-connected space lifts after fixing a preimage of one point. In this case the lift can be constructed by lifting paths from 0; homotopy invariance of path lifting and simple connectedness of $\mathbb{C}$ make the endpoint independent of the chosen path. Local inverses of q' are holomorphic, so the lift is holomorphic.

For $\omega \in \Lambda$,

$$q'(F(z+\omega)) = f(q(z+\omega)) = f(q(z)) = q'(F(z)).$$

Hence $F(z+\omega) - F(z) \in \Lambda'$. This difference is continuous on the connected plane and has values in a discrete set, so it is constant in z. Differentiation yields $F'(z+\omega) = F'(z)$. The continuous function F' is bounded on a closed fundamental parallelogram for Λ; periodicity then bounds it on all of $\mathbb{C}$. Liouville's theorem says a bounded entire function is constant. Thus $F' = c$ and $F(z) = cz + t$. The preceding differences now equal $c\omega$, proving $c\Lambda \subseteq \Lambda'$.

Conversely, this inclusion gives independence of representatives, and $cz + t$ is holomorphic in local coordinates. Two lifts of the same map differ by a constant lattice element, so their derivatives agree and their constants are congruent modulo Λ'. Finally, $f(0) = 0$ implies $t \in \Lambda'$; subtracting that element makes $t = 0$, and the resulting formula is additive. □

Corollary 6.2. *The tori $\mathbb{C}/\Lambda$ and $\mathbb{C}/\Lambda'$ are biholomorphic if and only if $c\Lambda = \Lambda'$ for some $c \in \mathbb{C}^\times$. A nonconstant holomorphic group homomorphism is a finite covering of degree $[\Lambda' : c\Lambda]$.*

Proof. Translations are biholomorphic, so an isomorphism can be made to preserve zero. By Theorem 6.1, it and its inverse have slopes c, d with $c\Lambda \subseteq \Lambda'$ and $d\Lambda' \subseteq \Lambda$. The identity composite has slope 1, so $dc = 1$. Multiplying the second inclusion by c gives $\Lambda' \subseteq c\Lambda$, proving equality. The converse and the degree statement follow from Proposition 5.2 applied to multiplication by c. □

Such a nonconstant group homomorphism is called an *isogeny*. If $\mathrm{Area}(\Lambda)$ is the Euclidean area of a fundamental parallelogram, then its degree also equals

$$[\Lambda' : c\Lambda] = \frac{|c|^2 \,\mathrm{Area}(\Lambda)}{\mathrm{Area}(\Lambda')}. \tag{6.2}$$

Indeed, multiplication by c has real determinant $|c|^2$, and a sublattice of index m has fundamental area m times that of the containing lattice. For the latter assertion, choose m coset representatives for the sublattice in the containing lattice. The union of the corresponding translates of a half-open fundamental parallelogram for the containing lattice is a fundamental region for the sublattice. The translates have disjoint interiors, so their areas add; boundary sets have area zero. In (5.1), $\mathrm{Area}(\Lambda_\tau) = \operatorname{Im}\tau$.

7. Deriving the modular equivalence criterion

Theorem 7.1. *For $\tau, \sigma \in \mathbb{H}$, the complex tori E_τ, E_σ are biholomorphic if and only if*

$$\sigma = \frac{a\tau + b}{c\tau + d} \quad \text{for some} \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2,\mathbb{Z}). \tag{7.1}$$

Their isomorphism classes are therefore parametrized by the orbit set $\mathrm{SL}(2,\mathbb{Z})\backslash\mathbb{H}$.

Proof. Suppose (7.1) holds and put $r = c\tau + d \neq 0$. The ordered pair

$$\omega_1' = d + c\tau = r, \qquad \omega_2' = b + a\tau = r\sigma$$

is a basis of Λ_τ: its integer change-of-basis matrix, relative to $(1, \tau)$, is $\left(\begin{smallmatrix} d & b \\ c & a \end{smallmatrix}\right)$ of determinant $da - bc = 1$. Hence $r\Lambda_\sigma = \Lambda_\tau$. Multiplication by r^{-1} induces a biholomorphism $E_\tau \to E_\sigma$.

Conversely, Corollary 6.2 gives $r\Lambda_\sigma = \Lambda_\tau$ for some nonzero r. The ordered basis $(r, r\sigma)$ is positively oriented because its area is $|r|^2 \operatorname{Im}\sigma > 0$. Write

$$r = d + c\tau, \qquad r\sigma = b + a\tau, \qquad a, b, c, d \in \mathbb{Z}.$$

As these vectors form a positively oriented lattice basis, the change-of-basis matrix has determinant 1. Dividing the second equality by the first proves (7.1). □

The imaginary part transformation is worth deriving rather than memorizing. For real a, b, c, d with $ad - bc = 1$,

$$\frac{a\tau + b}{c\tau + d} - \frac{a\overline{\tau} + b}{c\overline{\tau} + d} = \frac{(ad - bc)(\tau - \overline{\tau})}{|c\tau + d|^2}.$$

Thus

$$\operatorname{Im}\left(\frac{a\tau + b}{c\tau + d}\right) = \frac{\operatorname{Im}\tau}{|c\tau + d|^2} > 0. \tag{7.2}$$

The action preserves $\mathbb{H}$. Two familiar matrices give $\tau \mapsto \tau + 1$ and $\tau \mapsto -1/\tau$. The matrices I and $-I$ act identically on $\mathbb{H}$, so the effective group acting on this parameter space is $\mathrm{PSL}(2,\mathbb{Z})$. The displayed orbit set refers to isomorphism classes; it does not retain automorphism groups as a moduli stack would.

A marking here is a chosen positive ordered basis of the integral lattice (equivalently, of the first homology). Before changing that basis, τ is an actual parameter in $\mathbb{H}$. Forgetting the marking permits the modular identifications in (7.1). The distinction is not between an object and a different name for it: it is between two different amounts of specified data.

8. The square torus and a rectangular torus

Consider the same smooth torus $T = \mathbb{R}^2/\mathbb{Z}^2$, with the two constant structures

$$J_i = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad J_{2i} = \begin{pmatrix} 0 & -2 \\ 1/2 & 0 \end{pmatrix}.$$

Their complex coordinate lattices are $\Lambda_i = \mathbb{Z} + i\mathbb{Z}$ and $\Lambda_{2i} = \mathbb{Z} + 2i\mathbb{Z}$.

Proposition 8.1. *The complex tori E_i and E_{2i} are not biholomorphic, but are joined by degree-two isogenies in both directions.*

Proof. If they were biholomorphic, Theorem 7.1 would give $2i = (ai + b)/(ci + d)$ with integers a, b, c, d and $ad - bc = 1$. Taking imaginary parts by (7.2) gives

$$2 = \frac{1}{c^2 + d^2}.$$

The nonzero integer $c^2 + d^2$ cannot equal $1/2$, a contradiction.

For isogenies, define

$$\phi : E_{2i} \to E_i, \quad [z] \mapsto [z], \qquad \psi : E_i \to E_{2i}, \quad [z] \mapsto [2z].$$

The first is well-defined because $\Lambda_{2i} \subset \Lambda_i$; the second because $2\Lambda_i \subset \Lambda_{2i}$. Their kernels are, respectively,

$$\ker \phi = \{[0], [i]\} \subset E_{2i}, \qquad \ker \psi = \{[0], [1/2]\} \subset E_i.$$

Both have order two, so both maps have degree two. Each composite is the multiplication-by-two map on its source. Its kernel has four elements, as expected from the product of the degrees. □

In the fixed real coordinates on T, these maps are

$$\phi : \ [x, y] \longmapsto [x, 2y], \qquad \psi : \ [x, y] \longmapsto [2x, y].$$

The source of ϕ has structure J_{2i} and its target has structure J_i; for ψ the roles are reversed. Their integer matrices intertwine the appropriate structures, but have determinant 2, not 1. By contrast, the identity of the smooth torus is a diffeomorphism but is not holomorphic between these two structures. The identity lift in complex coordinates is

$$x + iy \longmapsto x + 2iy = \tfrac{3}{2}z - \tfrac{1}{2}\bar{z}.$$

There is no contradiction between a shared underlying smooth torus and Proposition 8.1: the shared smooth structure is weaker data.

More generally, E_i and E_{ni} are not biholomorphic for any integer $n > 1$, since (7.2) would imply $c^2 + d^2 = 1/n$. They nevertheless have degree-n isogenies $[z] \mapsto [z]$ in one direction and $[z] \mapsto [nz]$ in the other.

9. Measuring the failure of holomorphicity

For a differentiable map of planar complex coordinates, define

$$\partial_z = \tfrac{1}{2}(\partial_x - i\partial_y), \qquad \partial_{\bar{z}} = \tfrac{1}{2}(\partial_x + i\partial_y).$$

For the coordinate expression (3.3) of the underlying identity, denoted f_λ, we obtain

$$(f_\lambda)_z = \frac{1 + \lambda}{2}, \qquad (f_\lambda)_{\bar{z}} = \frac{1 - \lambda}{2}.$$

Its *Beltrami coefficient* is therefore

$$\mu_\lambda = \frac{(f_\lambda)_{\bar{z}}}{(f_\lambda)_z} = \frac{1 - \lambda}{1 + \lambda}, \qquad |\mu_\lambda| < 1. \tag{9.1}$$

The coefficient is zero exactly when the map is holomorphic. This calculation uses the source coordinate F_0 and the target coordinate F_λ; it does not change either complex structure.

With Euclidean metrics in these two coordinate planes, the real matrix of f_λ is $\operatorname{diag}(1, \lambda)$. Its largest to smallest singular-value ratio, the linear quasiconformal dilatation, is

$$K(f_\lambda) = \max\{\lambda, \lambda^{-1}\} = \frac{1 + |\mu_\lambda|}{1 - |\mu_\lambda|}. \tag{9.2}$$

For $\lambda \geq 1$, substitute $|\mu_\lambda| = (\lambda - 1)/(\lambda + 1)$ to obtain $K = \lambda$; for $0 < \lambda \leq 1$, substitute $(1 - \lambda)/(1 + \lambda)$ to obtain $K = 1/\lambda$. The formula passes to the torus because its local expression is unchanged. This computes the distortion of a *specified* map; we make no claim here that it minimizes distortion among all maps in a homotopy class, or that it computes a Teichmüller distance.

For comparison, A_λ has the coordinate expression $z \mapsto \lambda z$ in (3.2), with Beltrami coefficient zero. Even when λ is an integer and A_λ induces an isogeny, this is a local statement about holomorphicity, not a statement that the isogeny is a bijection.

10. What the example contributes to an approach to IUT

The preceding results are classical and independent of IUT. Their role is to make four kinds of data separation explicit:

(1) A real vector space can be fixed while its complex scalar operation varies. The identity need not preserve that operation.
(2) A comparison can preserve complex structures after transporting them, without preserving a separately fixed lattice.
(3) A map can be holomorphic and surjective without being an isomorphism; the lattice index measures its degree.
(4) Forgetting a marking changes which objects are regarded as equivalent, and which comparisons are permitted.

In [1, §3], Mochizuki discusses real linear dilations and different complex structures as classical motivation for comparisons of arithmetic structures. The broader survey [2, §2] also develops Teichmüller-theoretic analogies. These references motivate our choice of example; none of their arithmetic assertions is needed to prove it.

The analogy has a definite boundary. An IUT Hodge theater is not a real plane with a matrix J, and the Θ-link is not the map A_λ. The arithmetic constructions involve additional Galois-theoretic, multiplicative, and reconstruction data. In particular, the statements proved here do not construct an IUT link, establish multiradiality, or yield the numerical estimates in the IUT papers.

A productive reading question is nevertheless supplied by the example: for each proposed comparison, what is its domain, what is its codomain, which structure is preserved, and which extra choices or structures are being forgotten? On the real torus these questions have distinct, computable answers. Keeping them distinct is the elementary lesson to carry into more advanced arithmetic geometry.

11. Exercises with complete solutions

Exercise 1. Compute all real linear maps B commuting with J_λ. Determine which of them are invertible.

Solution. Let $B = \left(\begin{smallmatrix} r & s \\ t & u \end{smallmatrix}\right)$. Multiplying on both sides of $BJ_\lambda = J_\lambda B$ gives

$$\begin{pmatrix} s/\lambda & -r\lambda \\ u/\lambda & -t\lambda \end{pmatrix} = \begin{pmatrix} -\lambda t & -\lambda u \\ r/\lambda & s/\lambda \end{pmatrix}.$$

Thus $u = r$ and $s = -\lambda^2 t$. Writing $a = r$ and $b = \lambda t$ gives

$$B = \begin{pmatrix} a & -b\lambda \\ b/\lambda & a \end{pmatrix} = aI + bJ_\lambda.$$

Its determinant is $a^2 + b^2$, so it is invertible exactly when $(a, b) \neq (0, 0)$. In the coordinate F_λ it is multiplication by $a + ib$. □

Exercise 2. Find the complex structure for $\tau = 1 + 2i$. Verify its square explicitly. Show that E_{1+2i} and E_{2i} are biholomorphic.

Solution. Theorem 4.1 gives

$$J_{1+2i} = \frac{1}{2}\begin{pmatrix} -1 & -5 \\ 1 & 1 \end{pmatrix}.$$

The square of the integer matrix is $\left(\begin{smallmatrix} -4 & 0 \\ 0 & -4 \end{smallmatrix}\right)$, so $J_{1+2i}^2 = -I$. The lattices are actually equal:

$$\mathbb{Z} + (1 + 2i)\mathbb{Z} = \mathbb{Z} + 2i\mathbb{Z},$$

since replacing the second generator by its difference with the first is an integer basis change. The identity on $\mathbb{C}$ induces the desired isomorphism. In fixed real coordinates its matrix from J_{1+2i} to J_{2i} is $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, because $x + (1 + 2i)y = (x + y) + 2iy$. It is not the underlying identity of $\mathbb{R}^2$. □

Exercise 3. For a fixed $\lambda > 0$, determine all real scalars s for which sI induces a holomorphic group endomorphism of $E_{i\lambda}$. For $s \neq 0$, compute its degree and kernel.

Solution. A scalar matrix commutes with every J, so holomorphicity on the plane is automatic. On the fixed real lattice $\mathbb{Z}^2$, descent requires $s\mathbb{Z}^2 \subseteq \mathbb{Z}^2$, which is equivalent to $s \in \mathbb{Z}$. If $s = n \neq 0$, the kernel is

$$\left(\frac{1}{n}\mathbb{Z}^2\right)/\mathbb{Z}^2 \cong (\mathbb{Z}/|n|\mathbb{Z})^2,$$

with n^2 elements. Thus the degree is n^2, also the determinant of nI. For $s = 0$ the map is constant and has no finite covering degree. □

Exercise 4. Show that all positive structures on the real plane are isomorphic, but not all structures on the fixed real torus $\mathbb{R}^2/\mathbb{Z}^2$ are biholomorphic. Explain why neither conclusion determines whether the underlying identity is holomorphic.

Solution. For any $\tau, \sigma \in \mathbb{H}$, the real linear map $F_\sigma^{-1} F_\tau$ is a complex linear isomorphism between J_τ and J_σ. It need not carry $\mathbb{Z}^2$ onto $\mathbb{Z}^2$. On the torus, Proposition 8.1 gives the counterexample E_i, E_{2i}. The underlying identity of the plane, or of the fixed smooth torus, is holomorphic exactly when $J_\tau = J_\sigma$; by the bijection in Theorem 4.1, this means $\tau = \sigma$. Distinct parameters may still define isomorphic unmarked tori, as Exercise 2 demonstrates, through a different map. □

Exercise 5. Compute the ring of origin-preserving holomorphic endomorphisms of E_i. Determine its units and the degree of each nonzero endomorphism.

Solution. By Theorem 6.1, an endomorphism is multiplication by a number c satisfying $c(\mathbb{Z} + i\mathbb{Z}) \subseteq \mathbb{Z} + i\mathbb{Z}$. Applying this to 1 forces $c = a + bi$ with $a, b \in \mathbb{Z}$. Conversely, every such c works, since $ci = -b + ai$ is also in the lattice. Addition and composition of endomorphisms correspond to addition and multiplication of these numbers, so $\operatorname{End}(E_i) = \mathbb{Z}[i]$. For $c \neq 0$, formula (6.2) gives degree $|c|^2 = a^2 + b^2$. An endomorphism is invertible exactly when this degree is one, so the units are $1, -1, i, -i$. □

Exercise 6. Compute the endomorphism ring and the origin-preserving automorphism group of E_{2i}. Compare them with those of E_i.

Solution. For multiplication by c to preserve $\mathbb{Z} + 2i\mathbb{Z}$, its value at 1 must have the form $c = a + 2bi$, where $a, b \in \mathbb{Z}$. Conversely,

$$c(2i) = -4b + 2ai \in \mathbb{Z} + 2i\mathbb{Z},$$

so every such number defines an endomorphism. Hence $\operatorname{End}(E_{2i}) = \mathbb{Z}[2i]$ as a subring of $\mathbb{C}$. Its nonzero endomorphisms have degree $a^2 + 4b^2$, by (6.2). Degree one forces $b = 0$ and $a = \pm 1$. Thus its origin-preserving automorphism group has two elements, whereas that of E_i has four. This supplies another proof that the two tori are not biholomorphic: any biholomorphism can be translated to preserve zero, and conjugation would then identify their origin-preserving automorphism groups. □

References

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